Kelvin's vortex rings as ring electromagnets

Can the engine's inductance and Biot–Savart code compute Kelvin's 1867 vortex atoms through his ring-electromagnet analogy? — CLAUDE.md §6 (Kelvin)

kronspice simulation report · 2026-11-07 · Γ = 1, R = 1

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Summary

Kelvin's 1867 "On Vortex Atoms" proposes that atoms are closed vortex rings in a perfect liquid, and notes that the liquid's velocity round such a ring is the magnetic field of the "corresponding ring electromagnet", with the ring's impulse equal to that electromagnet's magnetic moment (pp. 9–10). This example takes the analogy literally: the circulation Γ plays the current I, the velocity plays B/μ_0, and the ring's kinetic energy is $\frac{1}{2}\rho\Gamma^2 \cdot L/\mu_0$, with ρ the liquid density and L a self-inductance. The engine's inductance and Biot–Savart code (milestone M9i) then computes the vortex ring directly. It sits beside the charged ring of *kelvin_ring* on the electron test bench.

- **Ring speed from the inductance matches Kelvin's formula** to −0.97 %, −0.26 % and −0.07 % for cores a/R = 0.20, 0.10 and 0.05; the difference falls as $(a/R)^2$, the order of the terms the thin-ring formula drops.
- **The analogy's identities hold:** the flow at the ring's centre is 0.50000 = Γ/2R (as the field at a loop's centre is $\mu_0 I/2R$), and the impulse $\rho\Gamma\pi R^2$ is the counterpart of the moment $I\pi R^2 = 3.14159$.
- **Two coaxial rings leapfrog:** 21 passes in 60 time units, period 2.92, radii swinging 0.730 … 1.211, the pair advancing at 0.4990 against 0.3288 for a single ring.
- **Invariants hold:** impulse drift 2.3×10^{-5}, energy drift 2.1×10^{-5}, which also checks that the Biot–Savart velocities and the elliptic mutual inductance agree.
- **Same shape, different dynamics:** a vortex ring moves itself; a current loop with the identical field does not. The analogy is static.

The construction

1. Ring speed

The self-inductance of a ring conductor of radius R and round core radius a with uniform current is Maxwell's mutual inductance of coaxial circles (Treatise Art. 701) averaged over pairs of filaments in the core section (Gauss–Legendre in ρ^2, midpoint in φ). The ring's speed is U = dE/dP at fixed circulation and core volume (a^2R constant), with energy $E = \frac{1}{2}\rho\Gamma^2 L/\mu_0$ and impulse $P = \rho\Gamma\pi R^2$. It is compared with Kelvin's $U = \Gamma/4\pi R\cdot(\ln 8R/a - \frac{1}{4})$ (Vol. IV Art. 3, as quoted by Saffman §10.2).

2. Leapfrogging

Two coaxial rings of equal Γ (Helmholtz 1858; Tait's smoke rings, Kelvin pp. 2, 11–12) are integrated with RK4; each moves at its self speed plus the other ring's Biot–Savart flow at its centre line. The energy is the self terms plus the mutual-inductance interaction, so impulse and energy must stay constant.

3. Where the analogy stops

The flow and the field share their shape but not their dynamics; see below.

Results (Γ = 1, R = 1)

Self-inductance and ring speed

Core a/R	$L/\mu_0 R$ (numeric)	Thin ring $\ln 8R/a - 7/4$	U (numeric)	Kelvin's U	Difference
0.20	1.95837	1.93888	0.27100	0.27366	−0.97 %
0.10	2.63730	2.63203	0.32798	0.32882	−0.26 %
0.05	3.32624	3.32517	0.38369	0.38398	−0.07 %

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The numeric inductance approaches the thin-ring value ln(8R/a) − 7/4 as the core shrinks, and the speed difference falls as (a/R)².

Leapfrogging rings

Two rings of core 0.1, 0.6 apart, integrated for 60 time units:

Quantity	Value
Passes	21, period 2.92
Radii swing	0.730 … 1.211
Pair advance	0.4990 (single ring 0.3288)
Impulse drift	2.3×10^{-5}
Energy drift	2.1×10^{-5}

Same shape, different dynamics

A vortex ring travels because its vorticity is carried along by the flow it makes. A current loop with the identical magnetic field is not moved by that field: in Maxwell's equations the ring electromagnet stays put. A self-propelled toroidal *electromagnetic* structure has to come from the full field equations, not from this static analogy; the standard case is Hellwarth and Nouchi's flying doughnut (*em/doughnut.go*), which travels at c.

A vortex ring in a neutral liquid has no charge and no coupling to an external field, so the Zeeman bench of *kelvin_ring* cannot be applied to it until a charge–vortex coupling is specified.

Validation

- Speed error against Kelvin's thin-ring formula shrinks as a² (test in *main_test.go*).
- Centre-flow identity: 0.50000 = Γ/2R.
- Leapfrog invariants: impulse and energy constant to 2.3×10^{-5} and 2.1×10^{-5} over 60 time units.

Limits

- Inviscid, incompressible liquid; round cores with uniform vorticity; coaxial rings only.
- The analogy maps the shape of the flow onto a magnetic field; it carries no electromagnetic dynamics and no charge, so it says nothing about vortex atoms' spectra.

Reproducing the results

```
go run ./examples/vortex_ring
go test ./examples/vortex_ring
```

Sources: W. Thomson (Lord Kelvin), "On Vortex Atoms," Proc. R. Soc. Edinb. 6 (1867); Mathematical and Physical Papers Vol. IV (1910), Art. 1, pp. 1–12, Art. 3, p. 67; H. von Helmholtz, J. reine angew. Math. 55, 25 (1858); P. G. Saffman, Vortex Dynamics (1992), §10.2; J. C. Maxwell, Treatise, Vol. II, Art. 701.

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